

浙大 2012

11. 10, 3

5. $W = p_0 V_0 + 4RT_0 \ln 2$

$Q = \frac{15}{2} RT_0 + W$

$\Delta S = 7R \ln 2$

6. (1) 对称, 中心为暗

(2) 3383 nm

7. $\frac{1}{6} \pi r^2 k - \frac{1}{3} \pi r V B$

1. 略

2. $\frac{1}{2} c t^4 + V_0 t + X_0$

$V = \frac{1}{3} c t^3 + V_0$

3. 352.

4. $\frac{1}{2} m v^2 \left(\frac{r_1^4}{r_2^2} - r_1^2 \right)$

5. $\frac{m_0}{\phi a b (1 - \frac{v^2}{c^2})}$

6. 0.2125 m/s

7. $R V_0 \frac{1}{r^2}$

8. 1×10^6

9. $\frac{M_0 I}{2R} - \frac{M_0 I}{2\pi R}$

10. $\frac{1}{4} I_0$

12. --

二.

1. --

2. (1) $\sqrt{\frac{2g}{3L}} (1 - \cos \theta)$

(2) $\frac{1}{3} m \sqrt{\frac{2g}{3}} (1 - \cos \theta)$

3. $E = \frac{32}{\ln \frac{5}{2}} \frac{1}{0.035}$

$V = 32 \frac{\ln \frac{5}{2.5}}{\ln \frac{5}{2}}$

4. (1) $X = A \cos(500\pi t + \frac{\pi}{50} x + \frac{\pi}{4})$

(2) $X = A \cos(500\pi t + \frac{\pi}{4})$

$\dot{X} = -500\pi A \sin(500\pi t + \frac{\pi}{4})$